

Introduction to Artificial Intelligence (X_401089)

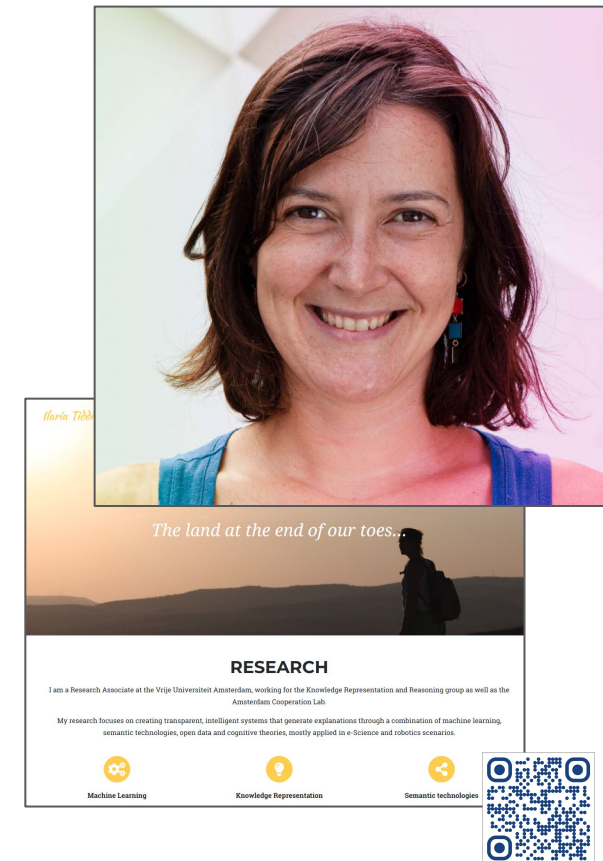
02-09-2024

Ilaria Tiddi

1. Welcome!

Quick round of presentations

- Ilaria Tiddi
 - Studies (computational) linguistics in Rome and Paris
 - PhD in Artificial Intelligence (UK)
-
- Assistant Prof. Hybrid Intelligence
 - Knowledge Representation and Reasoning, Knowledge Graphs, Explainable AI with KGs



<https://kmitd.github.io/ilaria/>

Quick round of presentations/Tutors

- ❖ [Stefan Schlobach](#), Group Head, Knowledge in AI
- ❖ [Frank Harmelen](#), Group Head, Learning and Reasoning

- ❖ [Victor de Boer](#), Associate Professor in User Centric Data Science
- ❖ [Michel Klein](#), Associate Professor in Social AI

- ❖ [Ilaria Tiddi](#), Assistant Professor in Knowledge in AI
- ❖ [Anil Yaman](#), Assistant Professor in Computational Intelligence
- ❖ [Karine Miras](#), Assistant Professor in Computational Intelligence

- ❖ [Vera N. Stebletsova](#), Lecturer in AI
- ❖ [Banno Postma](#), Lecturer in AI
- ❖ [Nimat Ullah](#), Lecturer in AI
- ❖ [Bob Borsboom](#), Lecturer in AI

Quick round of presentations/Guest Lecturers

❖ **Ilaria Tiddi**

- Finding Literature
- Basics for Computers
- Mathematics for AI
- Hybrid

Intelligence

❖ **Gusztai Eiben**

- Computational

Intelligence

❖ **Sara Lusini**

- Empirical

Evaluations

❖ **Koen Hindriks**

- Social Artificial

Intelligence

❖ **Stefan Schlobach**

- Knowledge Representation
- Ethical and Philosophical considerations related to AI

❖ **Benno Kruit**

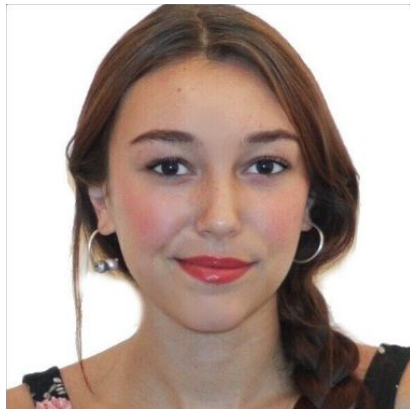
- Data Processing and Engineering

❖ **Peter Bloem**

- Machine Learning

Coordinator/Teaching Assistant of the Course & Communication Team

All communications: intro.to.ai.help.beta@vu.nl



Francesca Urgese
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AI & Cognitive Science
Master's



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2nd year AI Bachelor's



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2nd year AI Bachelor's

Teaching Assistants (working groups coordinators)



Francesca



Sena



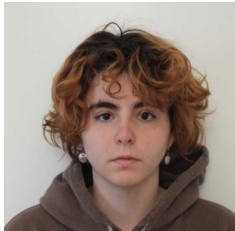
Ioana



Federico



Sanne



Ioana T.G.



Orkun



Noor



Damla



Sanskriti



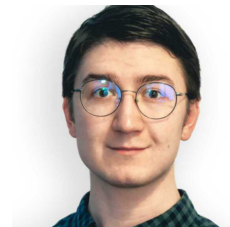
Joana



Corinna



Emir



Jan



Alisa

Quick round of presentations

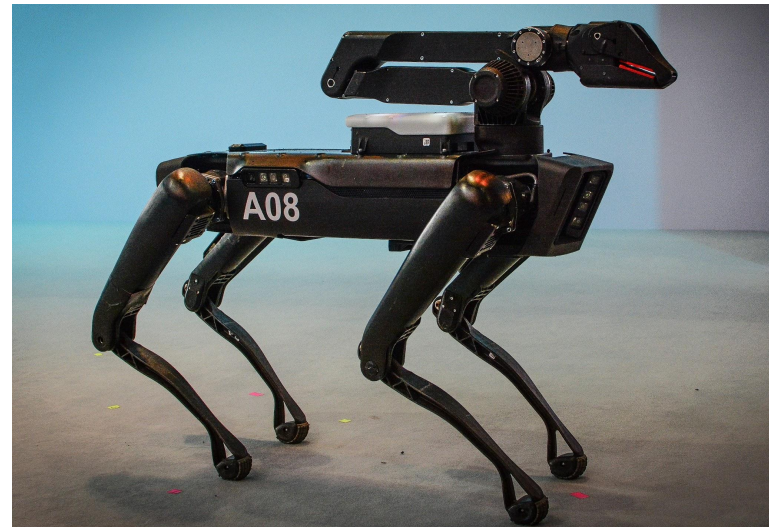
www.menti.com

use the code **8954 4332**



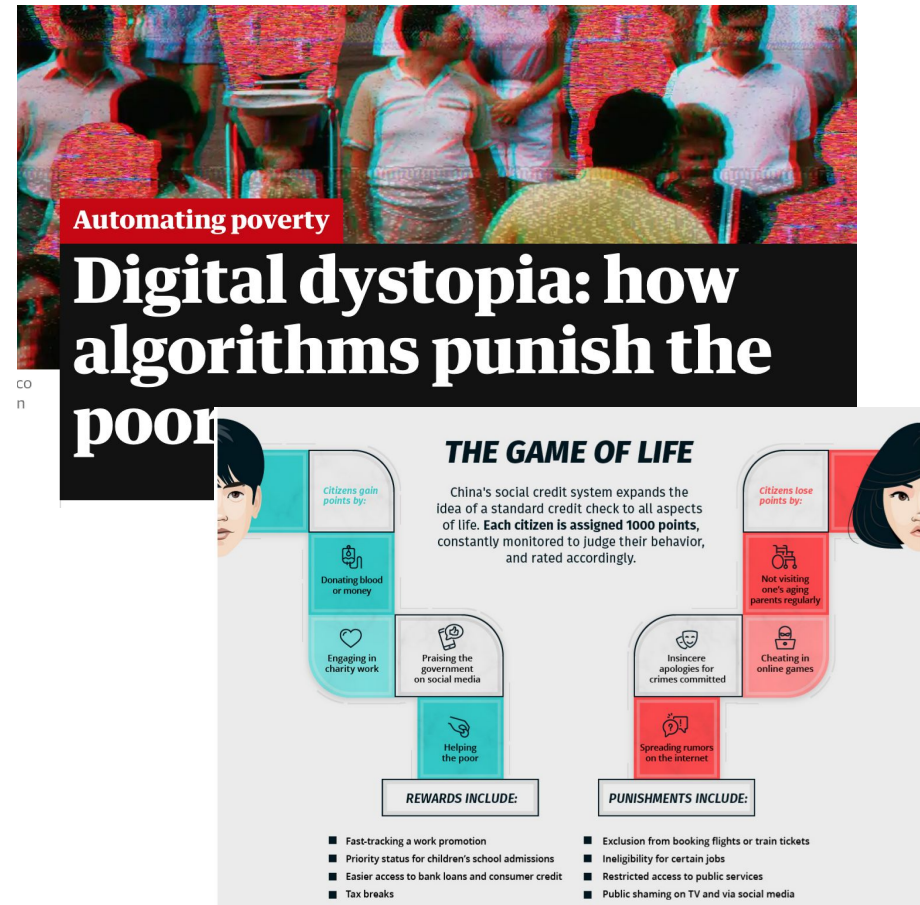
2. Why are we here?

Truth or sci-fi?



[1] <https://www.youtube.com/watch?v=fn3KWM1kuAw>

Truth or sci-fi?



[2] <https://www.youtube.com/watch?v=GQW6mUK2aDQ&list=PLJk1a21GKEZXxT9wwmnVle-zautJw4q12&index=2>

Truth or sci-fi?



What do you need to get to these technologies?

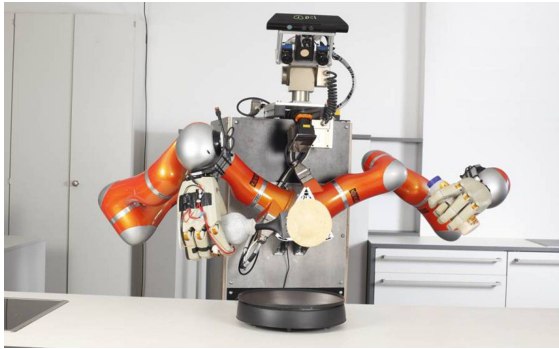
Main methods:

- Data Integration
- Knowledge Representation
- Automated Reasoning
- Learning from data (Machine Learning)
- Image processing
- Natural Language Processing (Speech & Understanding)

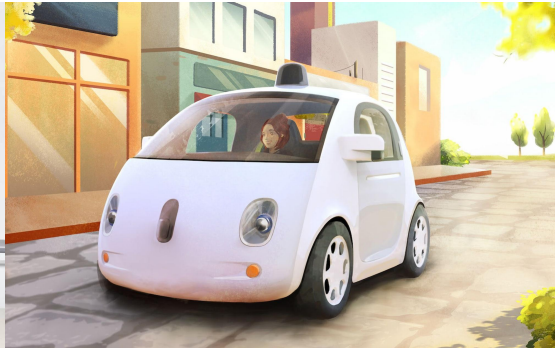
Overall approach:

optimising for the best expected outcome

The AI good



Highly-autonomous robots



Self-driving cars



Huge amounts of fast data

The AI bad

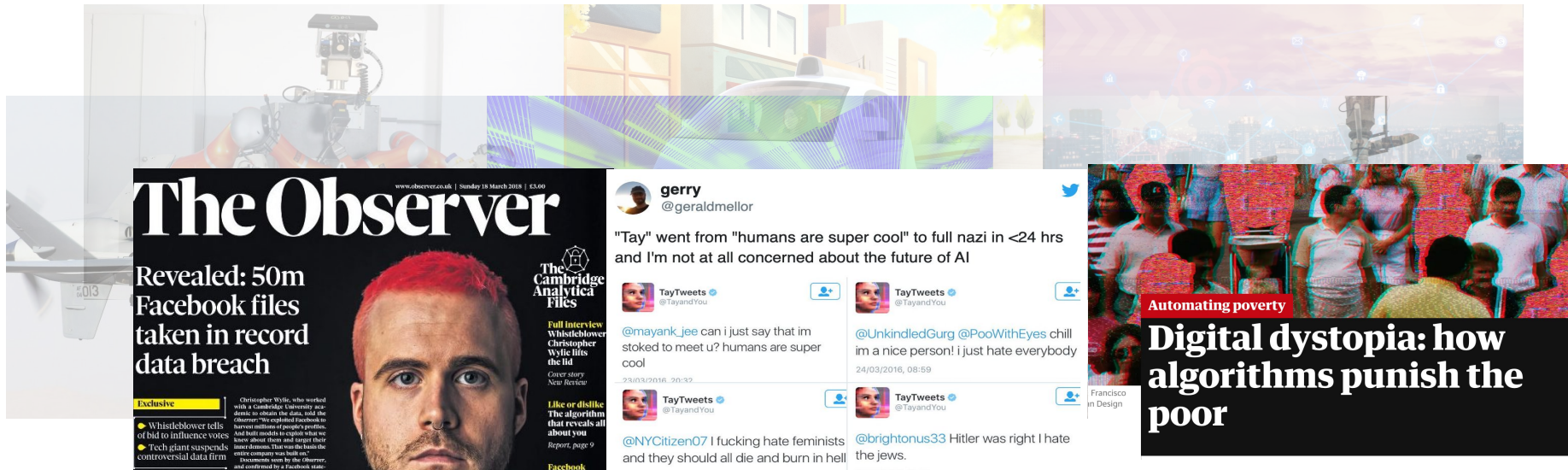


Military drones

Energy inefficiency

Security vs. surveillance

The AI ugly



Cambridge Analytica scandal

Unfair chatbots

Credit denial

3. AI at VU

What do we study?

Problem-driven Artificial Intelligence

The 4 steps

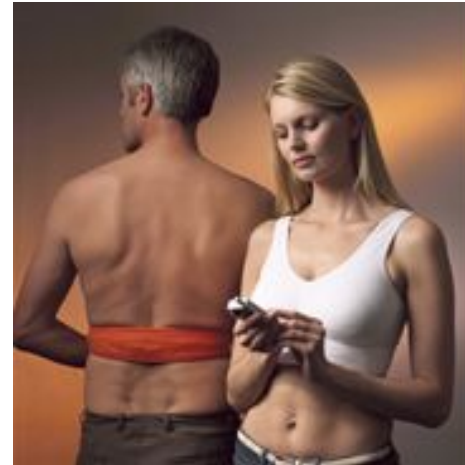
- **Identify** the problem
- **Understand** the context (theory)
- **Model** your solution
- **Realise** (application)

Examples

- How are you on time with heart problems?
- How do you help someone with a depression?
- How do you offer people security on the streets?
- How do you support lonely elderly people?

Daily Care for your Health

- Lead by Philips
- Methods for cardio-vascular diseases
- Sensors in clothing measure vital body functions
- Bad health? Doctor is warned!



Moodbuster

- Developed by Faculty of Psychology
- Therapy via Internet or Smartphone
- Measure state of mind and people's activities.
- Depressed? Tailor-made therapy is offered!



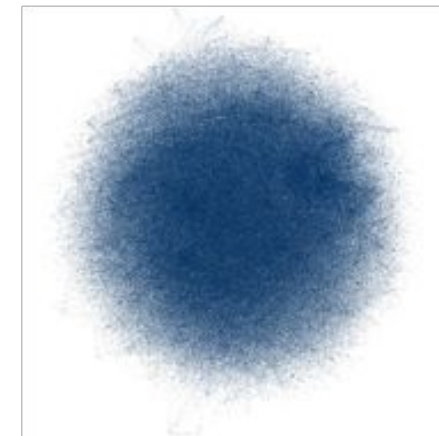
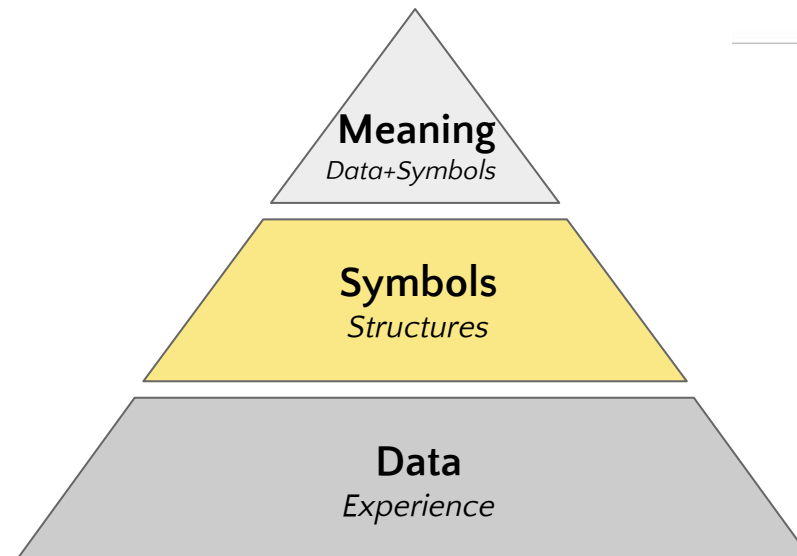
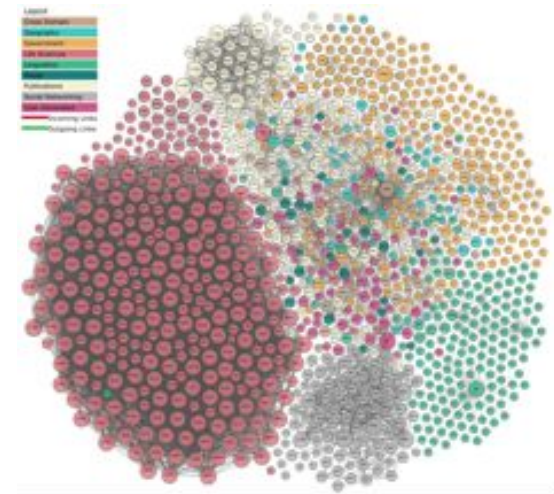
Robot companion

- Social Science and Social AI group
- Sensors measure interaction with 'owner'
- Robot shows emotional behaviour.
- Lonely? Robot will cheer you up!



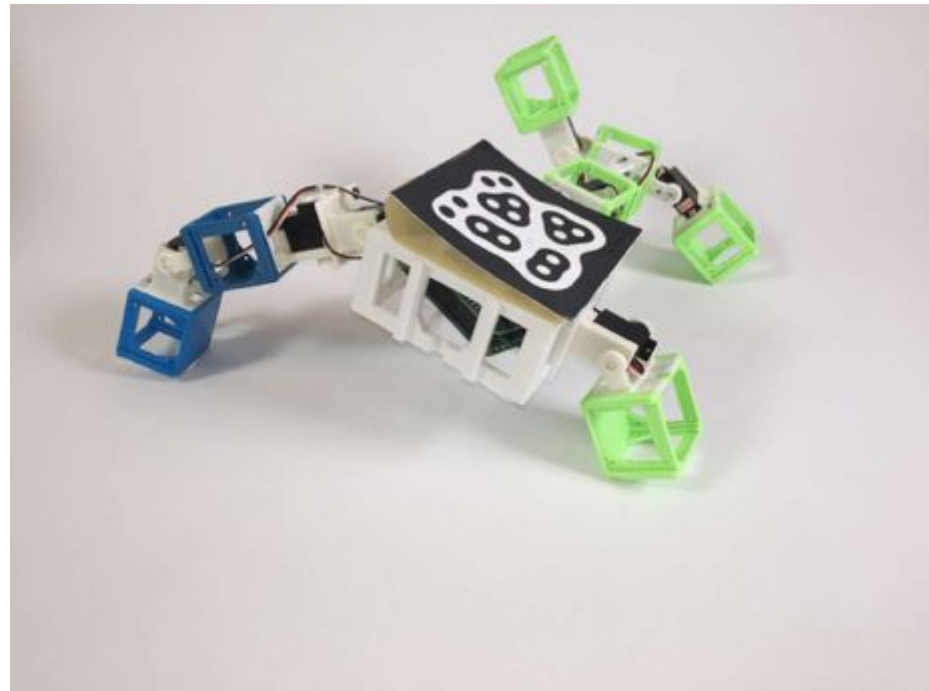
Building a Web of Knowledge

Knowledge Representation and Reasoning +
User Centred Data Science group

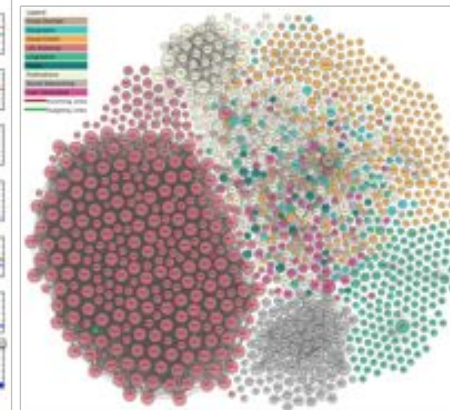
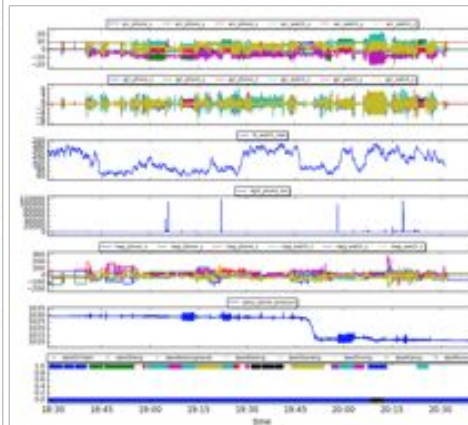
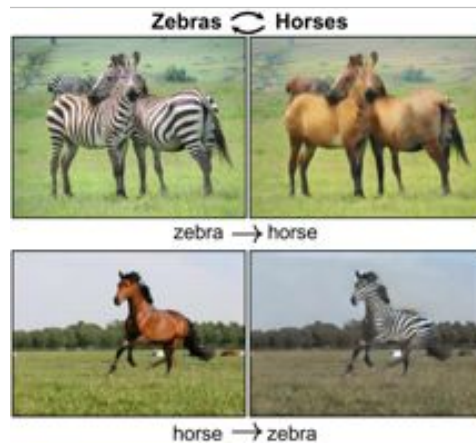


Evolution of the physical world

Computational Intelligence
group



From Artificial to Hybrid Intelligence



Main AI Methods

AI methods

- Dynamic Modelling
- Logical reasoning
- Knowledge representation
- Multi-Agent Systems
- Machine Learning
- Computational Intelligence

Support methods

- Programming
- Databases
- HCI

Academic Core

- Academic English
- Philosophy
- Statistical methods
- Linear algebra
- History of Science

BSc Artificial Intelligence Year 1

WEEK		36	37	38	39	40	41	42	43	44	45	46	47	48	49	#	51	52	1	2	3	4	5			
YEAR 1	period 1									period 2								period 3								
	Introduction to Psychology (6 ec) XB_0069									Intelligent Systems (6 ec) XB_0031								Project Intelligent Systems (6 ec) X_401076								
	Introduction to Python Programming (6 ec) XB_0082									XB_0082 (continued)																
	Introduction to Artificial Intellige... (3 ec) X_401089									Computational Thinking (3 ec) X_400475																
	English Language Test (0 ec) VU_ELT																									
		6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	33
	period 4									period 5								period 6								
	Dynamic Modelling for Hum-Cent Sys (6 ec) XB_0036									Human-Computer Interaction (6 ec) XB_0013								Information Management (6 ec) X_401085								
	Logic and Sets for AI (6 ec) XB_0083									History of AI (3 ec) XB_0113								Applied Programming ... (6 ec) XB_0102								
										Academic Writing (BETA) (3 ec) L_ETBAALG006																
										L_ETBAALG006 (continued)																

Year 2: Track Intelligent Systems

Interested in algorithmics of AI?

Intelligent Systems

period 1	period 2	period 3
Knowledge and Data (6 ec) X_400083	Statistical Methods for AI (6 ec) XB_0080	Project Conversa ... *** (6 ec) XB_0101
Data Structures and Algorithms AI *** (6 ec) XB_0043	Multi-Agent Systems (6 ec) XB_0024	
period 4	period 5	period 6
Linear Algebra for AI (6 ec) XB_0081	Databases (6 ec) X_401008	Project Collective Int ... *** (6 ec) XB_0026
Machine Learning (6 ec) X_400154	Computational Intelligence *** (6 ec) XB_0025	

Year 2: Track Socially Aware Computing

Interested in using AI to help people's lives?

Socially Aware Computing

period 1

Knowledge and Data (6 ec) X_400083
Robot Interaction **** (6 ec) XB_0040

period 2

Statistical Methods for AI (6 ec) XB_0080
Multi-Agent Systems (6 ec) XB_0024

period 3

Project Socially Aware...**** (6 ec) XB_0042
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period 4

Linear Algebra for AI (6 ec) XB_0081
Machine Learning (6 ec) X_400154

period 5

Databases (6 ec) X_401008	X_401008 (continued)
AI and Law **** (6 ec) XB_0070	XB_0070 (continued)
AI in Health **** (6 ec) XB_0020	XB_0020 (continued)

period 6

Project AI in Health **** (6 ec) XB_0021
The Law of AI **** (6 ec) XB_0084

BSc Artificial Intelligence Year 3

period 1

Free Choice: you can choose a minor or study abroad. Please check for information VU.nl.

period 2

Free Choice (continued)

period 3

Free Choice (continued)

period 4

Text Mining for AI
(6 ec) XB_0085

Automata and Complexity **
(6 ec) X_401049 (for track IS)

Cognitive Psychology for AI ***
(6 ec) XB_0072 (for track SAC)

period 5

Bachelor Project Artificial Intelligence
(15 ec) XB_40002

Philosophy and Ethics
(3 ec) X_400433

X_400433 (continued)

period 6

Get in touch with your academic advisor

Book an appointment through the website if:

- you'd like to discuss personal circumstances
- you need help with planning your studies
- anything remains unclear



Address:

NU building, room: 12A-45

E-mail:

academicadvice.cs.beta@vu.nl

Phone: 020 598 6718

To make an appointment:

[VUweb academic advice](#)

Departmental Honor Programme

- For 2nd year bachelor students AI & CS
- <https://vu.nl/en/education/more-about/application-honours-programme>
- <https://canvas.vu.nl/courses/65035>
- Not compulsory, but it is good for you CV
- Provisional entry (P1, P2 , results > 7.5)
- During 1st semester, 30 credits min.

Deadline: May 1st

MELD JE AAN ALS ***NAAM OPLEIDING*** AMBASSADEUR

Wat ga je doen?

- Studiekeizers informeren over jouw opleiding op een Bachelordag of meeloopdag
- Je neemt een scholier mee naar een college
- Eventueel: presentaties geven op scholen als je ouderejaars student bent

Why?

Flexibel: je kiest je eigen werkzaamheden en uren

Vergoeding: onder 21 jaar = €10 per uur en 21+ jaar = €12,50 per uur

Leuk?

Meld je aan! <https://vu.nl/nl/onderwijs/meer-over/ambassadeur-worden>

Sign Up for becoming an AI Ambassadeur

What are you going to do?

- Informing prospective students about your study programme during open days and Taster Days
- Taking a high school student with you to a lecture
- Giving presentations at schools (when you are a senior student)

Why?

Flexible: you choose your own hours and working activities

Reimbursement: Under 21 years € 10 per hour and 21+ years € 12.50 per hour

Nice?

Sign up! <https://vu.nl/en/education/more-about/become-an-ambassador>

4. Course practicalities

The Introduction to AI course

Main Goal

- What is Artificial Intelligence ?
- Develop academic skills
- Overview of the rest of your study

Digital education system: canvas.vu.nl

The course

Everything *is* on Canvas! <https://canvas.vu.nl/courses/78265>

Course summary:

Date	Details	Due
Fri, 13 Sep 2024	 Research Problem (groups assignment)	due by 23:59
Fri, 20 Sep 2024	 Literature (group assignment)	due by 23:59
Fri, 27 Sep 2024	 Approach (group assignment)	due by 23:59
Mon, 30 Sep 2024	 Basics of Computer	due by 23:59
Fri, 4 Oct 2024	 Experimental Evaluation (group assignment)	due by 23:59
Fri, 11 Oct 2024	 Poster (group assignment)	due by 23:59
Fri, 18 Oct 2024	 Video (group assignment)	due by 23:59
Thu, 31 Oct 2024	 Final report (group assignment)	due by 23:59
	 Diversity	
	 Exam Grade	

Date	Location	Type	Lecture	Teacher
Monday, Sept. 2 nd 13.30-15.00	HG-KC07 (330/82T)		Introduction to the course	Ilaria Tiddi
Friday, Sept. 6 th 15.30-17.00	HG-KC07 (330/82T)	Research Methodology	Finding Literature	Ilaria Tiddi
Monday, Sept. 9 th 13.30-15.00	HG-KC07 (330/82T)	Research Methodology	Group Projects and Team Working	Selin Çakmak
Friday, Sept. 13 th 15.30-17.00	HG-KC07 (330/82T)	Guest Lecture	Knowledge Representation	Stefan Schlobach
Monday, Sept. 16 th 13.30-15.00	HG-KC07 (330/82T)	Research Methodology	Basics of Computers	Ilaria Tiddi
Friday, Sept. 20 th 15.30-17.00	HG-KC07 (330/82T)	Guest lecture	Machine Learning	Peter Bloem
Monday, Sept. 23 rd 13.30-15.00	HG-KC07 (330/82T)	Research Methodology	Data Processing and Engineering	Benno Kruit
Friday, Oct. 27 th 15.30-17.00	HG-KC07 (330/82T)	Research Methodology	Methods in AI	Stefan Schlobach
Monday, Sept. 30 th 13.30-15.00	HG-KC07 (330/82T)	Research Methodology	Empirical Evaluations	Sara Lusini
Friday, Oct. 4 th 15.30-17.00	HG-KC07 (330/82T)	Guest Lecture	Social Artificial Intelligence	Koen Hindriks
Monday, Oct. 7 th 13.30-15.00	HG-KC07 (330/82T)	Research Methodology	Computational Intelligence	Gustzi Eiben
Friday, Oct. 11 th 15.30-17.00	HG-KC07 (330/82T)	Guest lecture	Ethical and Philosophical Considerations	Stefan Schlobach
Monday, Oct. 14 th 13.30-15.00	HG-KC07 (330/82T)	Research Methodology	Mathematics for AI (Discrete)	Ilaria Tiddi
Friday, Oct. 18 th 15.30-17.00	HG-KC07 (330/82T)	Guest lecture	Hybrid Intelligence	Ilaria Tiddi

What will you do (Final Project)

Solving a real-world problem with AI

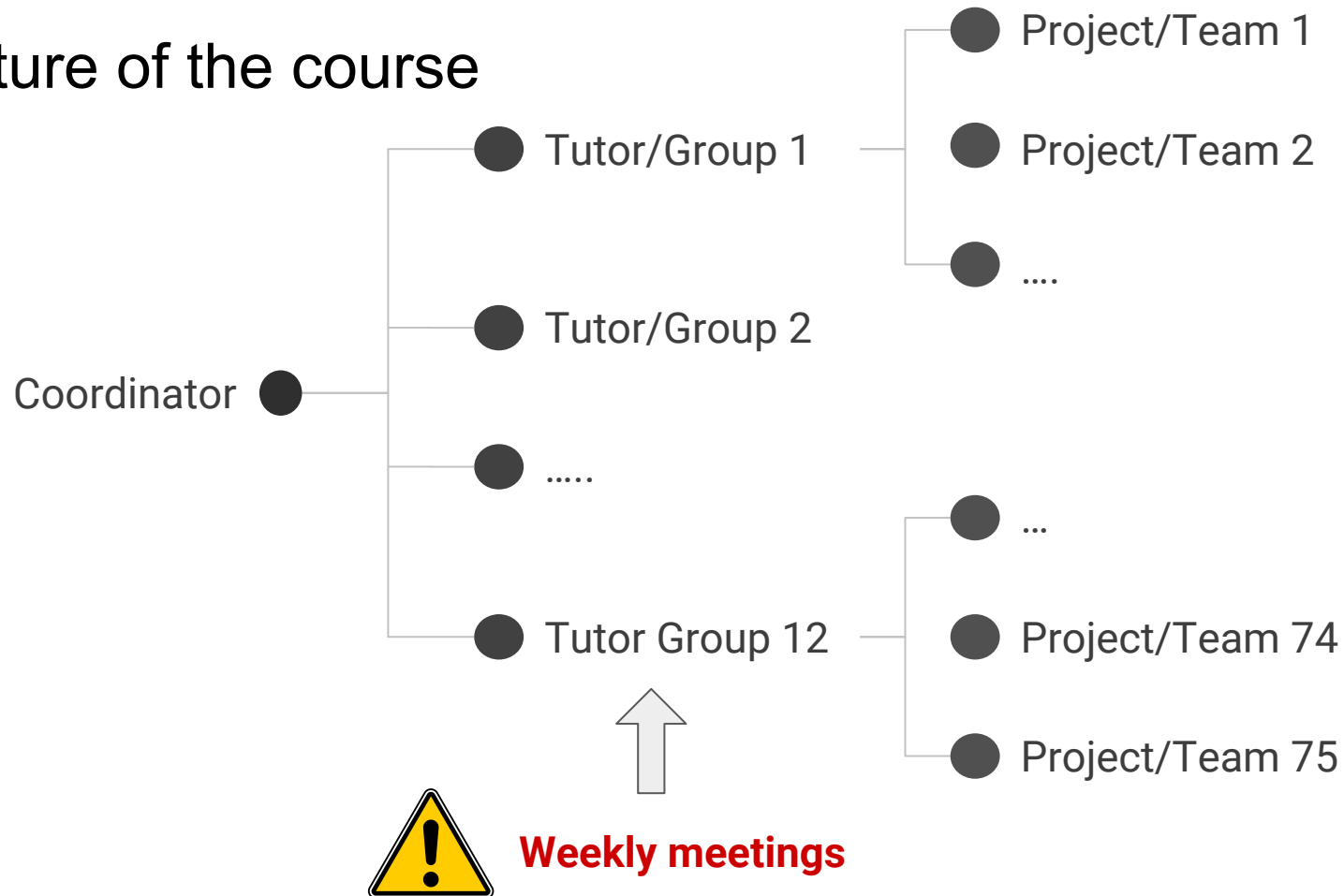
- Understand the problem
- Research state-of-the-art
- Find a computational AI solution
- Design an evaluation to validate your solution
- Report scientifically about your work
- Present your work
- Video + Poster

Win the competition: “Best Intro AI project” → cool prizes!

Structure of the course

- Lectures
 - 2 per week
 - Methodology lectures : how to do research in AI
 - Guest lectures : what types of research in AI exist
- Working groups
 - Tutoring sessions : weekly, monitoring progress
 - 1 tutor group: 5 Project groups (25 Students)
 - 1 Tutor group: 1 tutor and 2 Teaching Assistants
 - Home work and group work weekly (or daily :))

Structure of the course



Tutor working session

		Presence of TAs & Tutor			Preferable presence of TAs & Tutors			Preferable presence of TAs & Tutors
Deadline	Assignment	1st tutor session	2nd tutor session	3rd tutor session	4th tutor session	5th tutor session	6th tutor session	7th tutor session
13-Sept-24	Research Problem	Project Group formation +  Working on	Presentations					
20-Sept-24	Literature		 Instructions from TAs and working on it	Presentations				
27-Sept-24	Approach			 Instructions from TAs and working on it	Presentations			
04-Oct-24	Experimental Evaluation				 Instructions from TAs and working on it	Presentations		
11-Oct-24	Poster					 Instructions from TAs and working on it		
18-Oct-24	Video						Working on Video script	
31-Oct-24	Final Report							 Instructions from TAs and working on it

Communication

- All (any, stupid, insightful, cheerful) communications to :
intro.to.ai.help.beta@vu.nl
 - **Start the subject with : [IntroAI]**
- Slack : C07DREQD82Y
- Canvas
- Project questions? Contact your TA first
- In case of emergency : i.tiddi@vu.nl
 - Contact the TAs
 - Emergency = no answer within 1 day

The Lectures

2. Sept 13.30-15.00		Introduction to the course	Ilaria Tiddi
6. Sept 15.30-17.00	Research Methodology	Finding Literature	Ilaria Tiddi
9. Sept 13.30-15.00	Research Methodology	Group Projects and Team Working	Selin Çakmak
13. Sept 15.30-17.00	Guest lecture	Knowledge Representation	Stefan Schlobach
16 Sept 13.30-15.00	Research Methodology	Basics of Computers	Ilaria Tiddi
20 Sept 15.30-17.00	Guest lecture	Machine Learning	Peter Bloem
23 Sept 13.30-15.00	Research Methodology	Data processing and Engineering	Benno Kruit
27 Sept 15.30-17.00	Research Methodology	Methods in AI	Stefan Schlobach
30 Sept 13.30-15.00	Research Methodology	Empirical Evaluations	Sara Lusini
4 Oct 15.30-17.00	Guest lecture	Social Artificial Intelligence	Koen Hindriks
2 Oct 13.30-15.00	Research Methodology	Computational Intelligence	Gustzi Eiben
11 Oct 15.30-17.00	Guest lecture	Ethical and Philosophical considerations related to AI	Stefan Schlobach
14 Oct 13.30-15.00	Research Methodology	Mathematics for AI (Discrete)	Ilaria Tiddi
18 Oct 15.30-17.00	Guest lecture	Hybrid Intelligence	Ilaria Tiddi

The Tutoring/Mentoring groups

**Time to register
until Sept 9th!**

Tutor Group Number	Project Group Number	Day	Time	Location	Student Assistants	Tutor
1	1,2,3,4,5	Wednesday	13.30 - 15.15	NU-2B11 (24/12T)	Ioana S., Sena	Ullah, N. (Nimat)
2	6,7,8,9,10	Wednesday	17.30 - 19.15	HG-0G28 (25/12T)	Sanskriti, Corinna	Postma, B. (Banno)
3	11,12,13,14,15	Wednesday	17.30 - 19.15	NU-2B11 (24/12T)	Francesca, Damla	Harmelen, F.A.H. van (Frank)
4	16,17,18,19,20	Thursday	15.30 - 17.15	NU-6A52 (28/14T)	Jan, Emir	Tiddi, I. (Ilaria)
6	21,22,23,24,25	Wednesday	17.30 - 19.15	NU-6A52 (28/14T)	Alisa, Ioana S.	da Silva Miras de Araujo, K. (Karine)
7	26,27,28,29,30	Thursday	17.30 - 19.15	HG-0G28 (25/12T)	Joana, Sanne	Schlobach, K.S. (Stefan)
8	31,32,33,34,35	Thursday	17.30 - 19.15	NU-4B43 (34/17T)	Orkun, Noor	Yaman, A (Anil)
9	36,37,38,39,40	Wednesday	17.30 - 19.15	NU-4A67 (34/17T)	Federico, Jan	Klein, M.C.A. (Michel)
10	41,42,43,44,45	Wednesday	15.30 - 17.15	NU-6A52 (28/14T)	Federico, Joana	Stebletsova, V.N. (Vera)
11	46,47,48,49,50	Wednesday	15.30 - 17.15	HG-03A11 Agora 5 (30/15T)	Francesca, Sena	Borsboom, B. (Bob)
12	51,52,53,54,55	Thursday	15.30	HG-1G10 (25/12T)	Sanskriti, Ioana T.G.	Lisowska, A.J.

NB: group 5 has been closed!



Assignments

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Fri, 13 Sep 2024	 Research Problem (groups assignment)	due by 23:59
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	 Diversity	

Warning

Last Weeks :41,42,43

Less lectures, more time for:

- video
- poster
- Final report

Examples

You Dont Drink Alone

SO WHY QUIT ALONE?

Introducing PAAI: The AI powered personal coach

Tailored for you
Our algorithms make sure you get advice useful for your situation

Track your stats
Motivate yourself by tracking your stats and setting goals for yourself

Reach out to friends & family
We believe that your close ones are one of the most important factors for your progress, so get them involved with our nifty social features

Get inspired
The app provides you with motivating facts, health updates and plenty of other stuff to inspire you along your journey

Hi James

Did you know?
An estimated 50,000 people die from alcohol-related causes annually, making alcohol the third preventable cause of death in the United States.

Reach out
How do you feel?
Statistics

QUICK STATS

70 DAYS	14 URGES	€200 MONEY
SCORED	IGNORED	SAVED

GET IT ON
Google Play

PA/AI

Download on the
App Store

VU
Vrije Universiteit
 Brussel

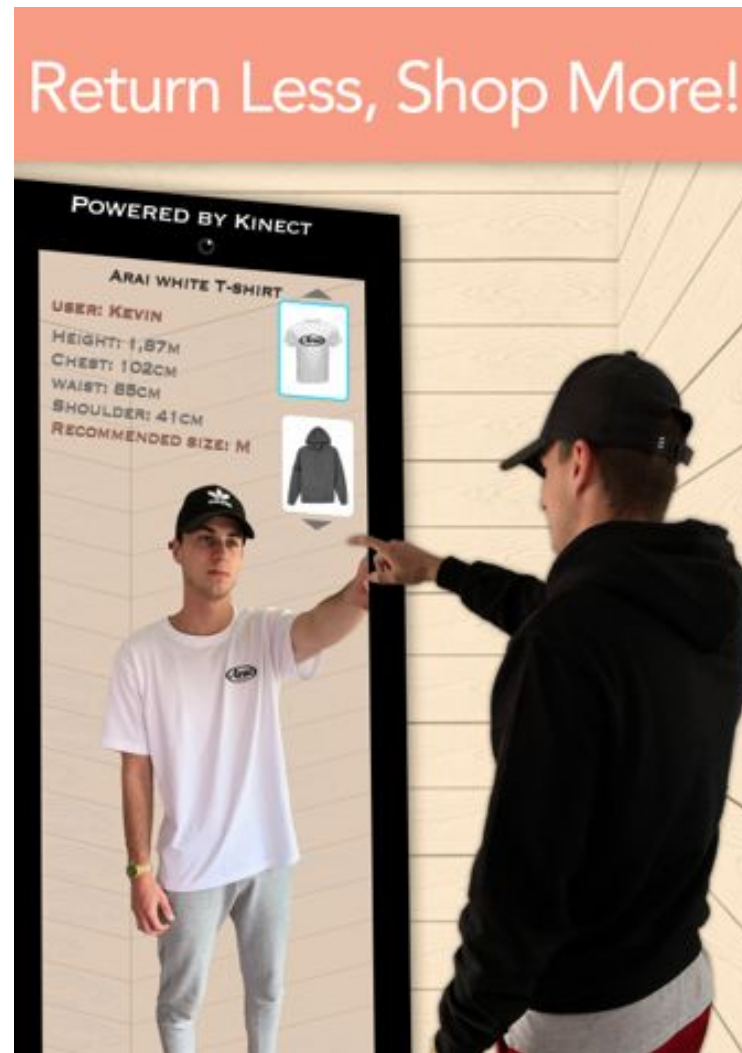
Examples



Examples



Examples



The bottom line...

Be creative!

Examples of projects, posters, videos :

<https://canvas.vu.nl/courses/78265/pages/examples-projects>

Canvas > Pages > Example-projects

Grading System

- 30% MC Exam
- 40% Group Project
- 5 % Poster
- 5 % Video
- 10% Diversity & Team Assignment
- 10 % Basics of Computer Assignment

This week tutoring meeting

Get to know each other

Come up with a number of possible ideas for your project

- *'fall detection at home',*
- *'app. to quit smoking',*
- *'smart car to assist driver'*

Describe the problem for each possibility, and an idea for a 'realistic' solution (in a few lines).

Present and discuss your ideas during the workgroup meeting.

That's it

Still questions?